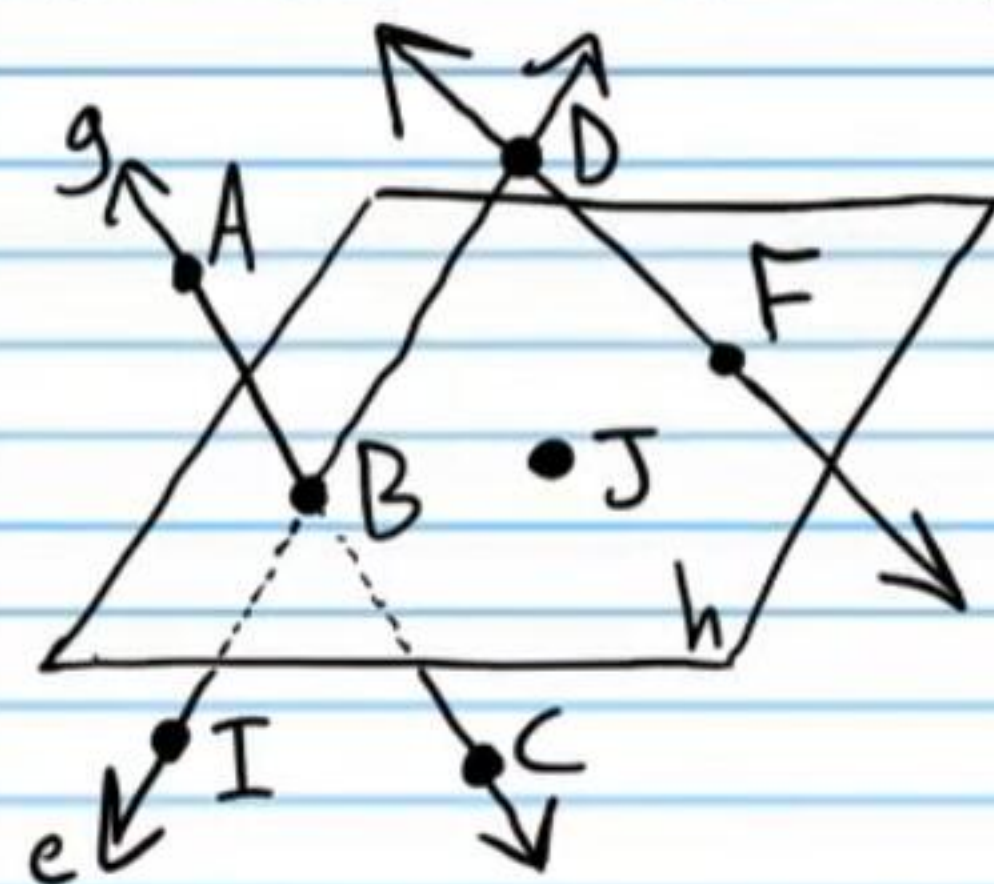


Fundamental Geometry (Part I) Homework

Name
Date
Course



Identify each of the following.

① F

② $\overleftrightarrow{IB}$

③ BDF

④ $\overline{BD}$

⑤ $\overleftrightarrow{AC}$

⑥ Name three collinear points.

⑦ What is another name for $\overleftrightarrow{AC}$?

⑧ What is the intersection of $\overleftrightarrow{AB}$ and $\overleftrightarrow{ID}$?

Answers

Point

Ray

Plane

Segment

Line

A, B, C,
or I, B, D

g, $\overleftrightarrow{AB}$, or $\overleftrightarrow{BC}$

B

⑨ Name a point that does not lie in plane h.

⑩ Name a point that is not collinear with $\overleftrightarrow{DF}$.

⑪ What is another name for $\overleftrightarrow{ID}$?

⑫ What is another name BTF?

⑬ Name two opposite rays.

⑭ Are A, B, and C collinear?

⑮ Name three points that are in plane h.

D, A, I, or C

A, B, I, C, or J

e, $\overleftrightarrow{IB}$, or $\overleftrightarrow{BD}$

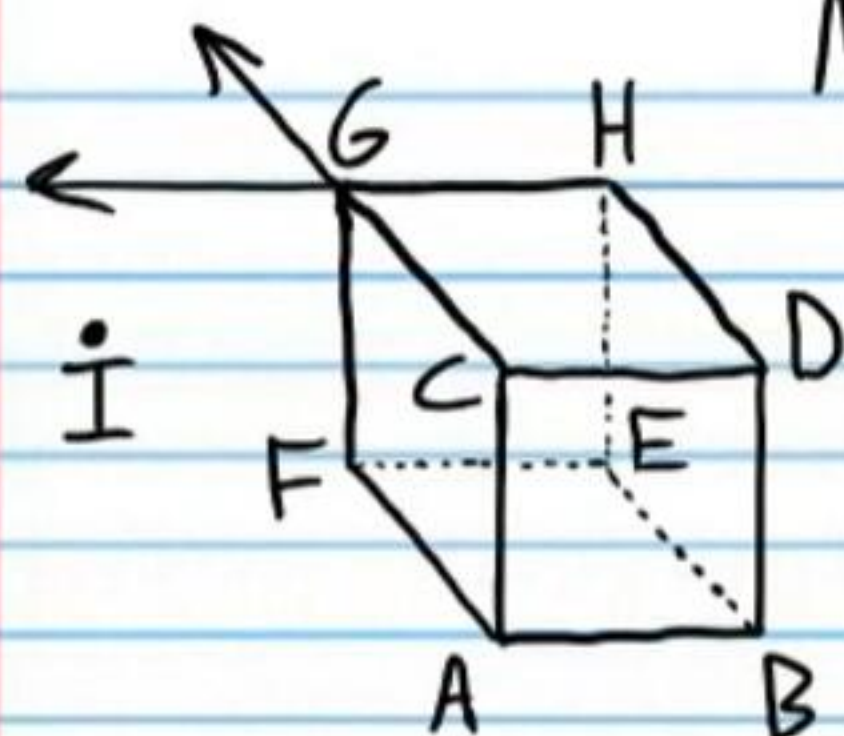
h

$\overleftrightarrow{BA}$ & $\overleftrightarrow{BC}$ or
 $\overleftrightarrow{BD}$ & $\overleftrightarrow{BI}$

Yes

B, J, and F

More Practice



16 Name the intersection of $\overline{ABE}$ and $\overline{DBC}$. Note: These are planes with finite length, which means that their intersection will be segments, not lines.

$\overline{AB}$

17 Name three planes that intersect at E.

$\overline{FEB}$, $\overline{BDH}$,
and $\overline{GHE}$.

18 Name the intersection of $\overrightarrow{HG}$ and $\overrightarrow{CG}$.

$\overline{G}$

19 Are F, B, and A collinear?

No

20 Name the intersection of $\overline{FBA}$ and $\overline{GHD}$.

No intersection

21 Are G, C, and D coplanar?

Any three points are coplanar.

22 Are F and A collinear?

Any two points are collinear.

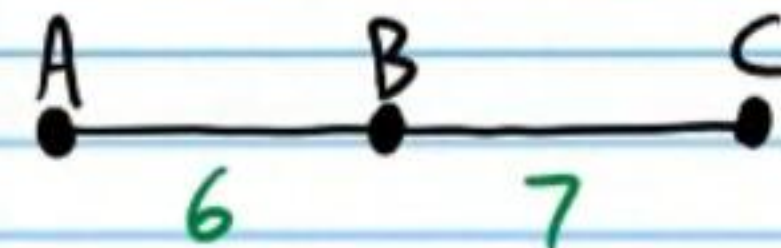
23 Are C and D coplanar?

Any two points are coplanar

24 Are G, H, and I collinear?

No

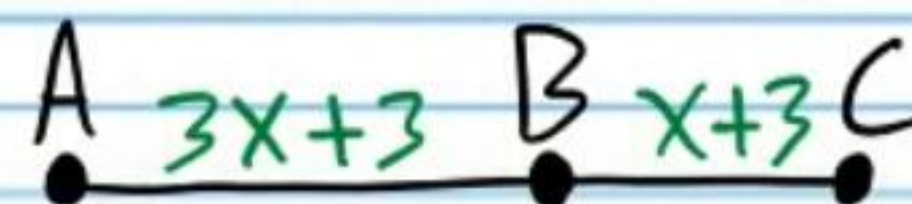
25 Find AC.



13

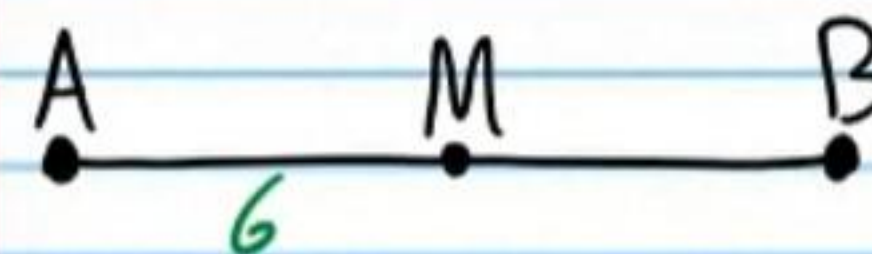
26 If $AC = 10$, find x.

$x = 1$



27 If M is the midpoint of $\overline{AB}$, find MB.

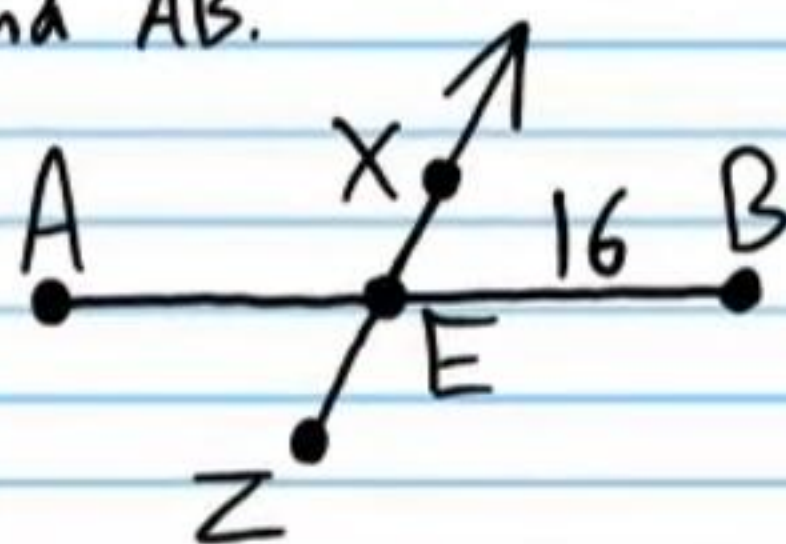
6



28 If $A(6,1)$ and $B(6,8)$ are endpoints of a segment, and $C(-1,-1)$ and $D(-1,6)$ are endpoints of another segment, are $\overline{AB}$ and $\overline{CD}$ congruent?

Yes

②9 If $\overrightarrow{ZX}$ bisects $\overline{AB}$,
Find AB .



32

③0 If L bisects $\overline{AB}$,
Find AB .



20

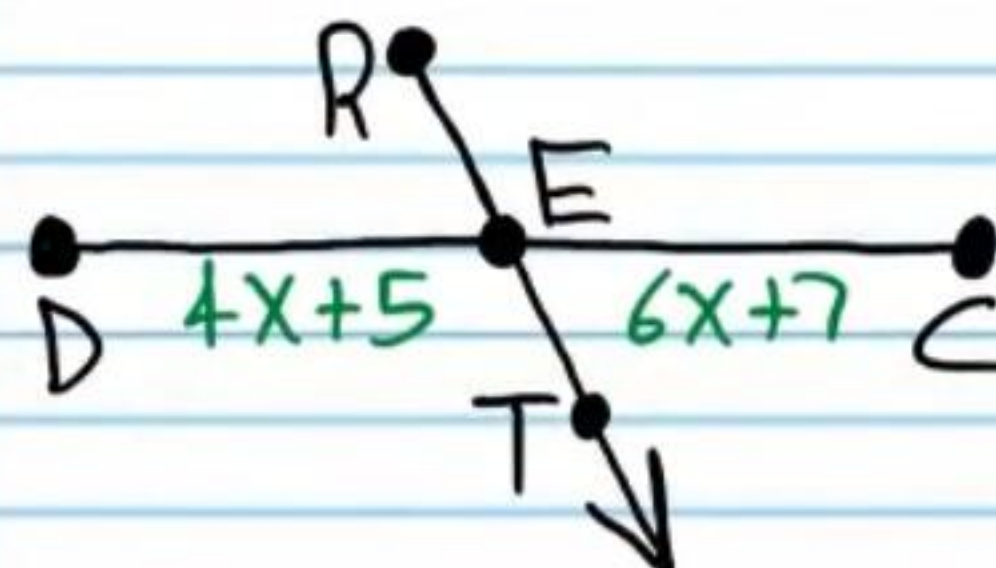
③1 Find the midpoint
of the segment with
endpoints $A(-4, 2)$ and $B(-4, 5)$.

$(-4, 3.5)$

③2 If $A(6, 7)$ is an endpoint
of the segment $\overline{AB}$, and
 $M(4, 7)$ is the segment's midpoint,
find the coordinates of B .

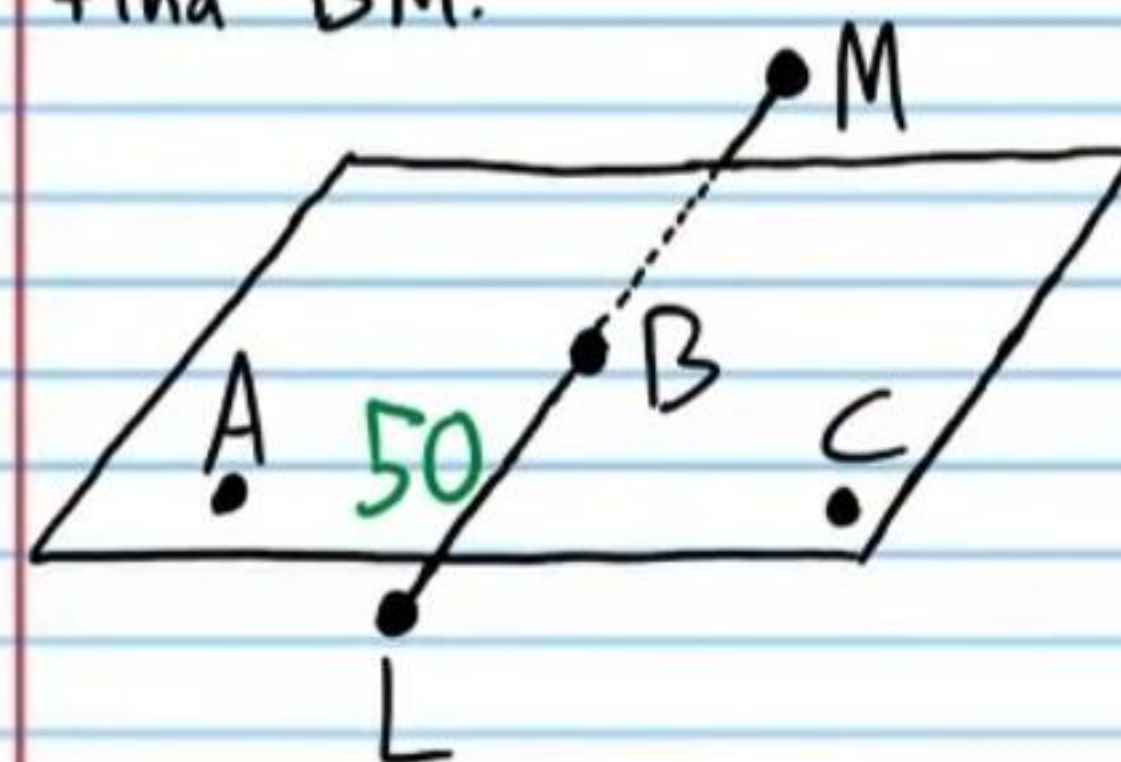
$B(2, 7)$

③3 If $\overrightarrow{RT}$ bisects $\overline{DC}$,
Find x .



$x = -1$

③4 If $\overline{ABC}$ bisects $\overline{LM}$,
Find BM .



50

35) If M is the midpoint of $\overline{AB}$,
Find the length of MB if
 $AM = -4x + 29$ and $MB = 8x + 5$.

21